0. Data Structures & Algorithms

"Data dominates. If you've chosen the right data structures and organized things well, the algorithms will almost always be self-evident."

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Algorithms are not just concepts used in computers; they already exist in our daily lives. Every time we follow a process step by step to achieve a result, we are essentially executing an algorithm.

- One of the simplest examples is a cooking recipe. A recipe provides a list of inputs (ingredients), a sequence of instructions, and a final output—the completed meal. When someone cooks a new dish, they carefully follow each instruction in order, and by executing those steps correctly, they obtain the intended result.

Many other activities around us also behave like algorithms.

- Daily routines, such as brushing your teeth, getting dressed, and preparing breakfast, follow a structured sequence designed to achieve a goal efficiently.

- Tying shoelaces follows a repeated pattern of steps that eventually secures the shoe properly.
- Modern traffic light systems rely on algorithms to process vehicle data from sensors and regulate traffic flow safely instead of changing lights randomly.
- Navigation and GPS applications such as Google Maps also use sophisticated routing algorithms to analyze traffic conditions, road closures, and distances in order to determine the fastest route to a destination.

Similarly, computers cannot simply “guess” what users want them to do. They require precise instructions written in a programming language. These instructions are organized into algorithms that guide the computer through a sequence of operations to produce a desired result. Without algorithms, computers would not be able to solve problems, process data, or perform tasks effectively.

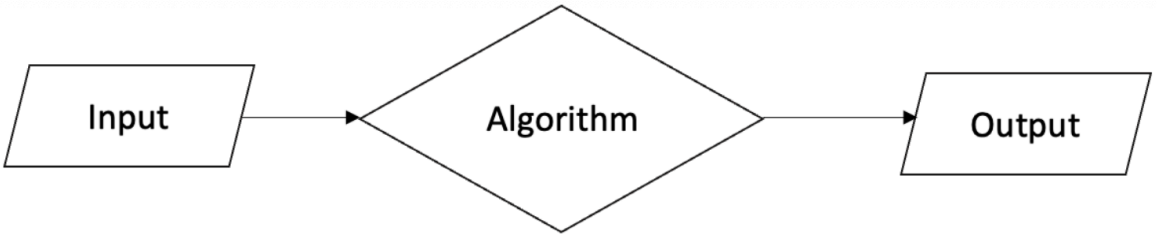
1. Algorithms

An algorithm is a sequence of well-defined instructions used to solve a problem or perform a computation. It accepts some form of input, processes it through a series of computational steps, and produces an output within a finite amount of time. In simpler terms, an algorithm transforms input into meaningful output through logical procedures.

Algorithms possess several important characteristics:

- First is **possibility**, meaning the steps can be explained clearly and implemented in a programming language.
- Second is **finiteness**, which means the algorithm must eventually terminate after a limited number of steps.
- Third is **determination**, where every step produces a predictable and unambiguous outcome.
- Fourth is **accuracy**, ensuring that valid inputs generate correct outputs.
- Finally, algorithms should also emphasize **efficiency**, aiming to save execution time and memory usage while maintaining a clean and effective design.

The general structure of an algorithm can be summarized as:



Input → Algorithm → Output

The algorithm itself may represent many different processes, such as searching, sorting, machine learning models, game strategies, numerical methods, or neural networks. As long as a process takes an input and produces an outcome, it can be considered an algorithm. Because of this, algorithms are fundamental in many fields including Computer Science, Artificial Intelligence, Data Science, Programming Languages, and Operations Research.

To better understand how algorithms work, let us examine a simple example: finding the largest number in a sequence. Suppose we are given the list:

```
[1, 4, 5, 6, 3, 10, 7]
```

The first step is to initialize a variable that stores the current maximum value. A common approach is to assign it the value `0` or the first element of the sequence.

For sequences that may contain negative numbers, programmers often initialize the value with negative infinity. In Python, this can be represented as:

```
float("-inf")
```

Next, the algorithm compares each element in the sequence with the current maximum value. If the current element is greater, the maximum value is updated.

- Starting with `x = 0`, the algorithm compares `1 > 0`, which is true, so `x = 1`.
- It then compares `4 > 1`, updating `x = 4`.
- The process continues:
  - `5 > 4` → `x = 5`

- `6 > 5` → `x = 6`
- `3 > 6` → False, so `x = 6`
- `10 > 6` → `x = 10`
- `7 > 10` → False, so `x = 10`

After all elements have been processed, the algorithm concludes that the largest value in the sequence is `10`.

An important observation is that for a sequence containing `N` elements, the algorithm performs `N - 1` comparisons. Each new element must be compared against the current maximum value, but the final maximum does not need to be compared against itself.

This example demonstrates how algorithms systematically solve problems through clear, logical, and finite steps.

2. Data Structures

Data structures are one of the core foundations of Computer Science and the central concept of this course. Even before formally learning these advanced data structures, you will constantly encounter them because algorithms cannot function without data to operate on.

Data itself can be understood as collections of values stored inside variables. Within these collections are smaller individual units called *data items* or *sub-items*. At a higher level of abstraction, these may also represent entities, instances, or objects belonging to a class.

A data structure is a specialized way of organizing, storing, and retrieving data inside a computer so that the information can be accessed and manipulated efficiently. Beyond machines, data structures also help humans better understand how information is arranged logically. In many ways, data structures are the “containers” that algorithms depend on to solve problems effectively.

Every data structure is generally evaluated through three major characteristics:

- **Correctness** refers to whether the data structure behaves accurately and reliably. Operations such as insertion, deletion, searching, or updating should always produce the expected result while properly handling unusual or edge-case situations.

- **Time Complexity** measures how efficient an operation is in terms of execution time as the input size grows. It helps programmers estimate how quickly a program can perform tasks when handling larger amounts of data.
- **Space Complexity** measures how much memory is required for a data structure or algorithm to function. Efficient programs aim to minimize unnecessary memory usage, especially when dealing with massive datasets or systems with limited resources.

Data structures are divided into two major categories: **primitive** and **non-primitive** data structures:

- Primitive data structures are the most basic units supported directly by programming languages. These include integers, floating-point numbers, characters, strings, and Boolean values. They are simple, lightweight, and serve as the building blocks for more advanced structures.
- Non-primitive data structures are built using primitive data types to organize larger and more complex collections of information. These structures support more advanced operations such as searching, sorting, insertion, deletion, and traversal. Non-primitive structures are further divided into **linear** and **non-linear** structures.
    - Linear data structures organize data sequentially, meaning elements are stored in a specific order. Examples range from simple structures such as lists, tuples, sets, and dictionaries to more advanced structures like arrays, tensors, linked lists, stacks, and queues.
    - Non-linear data structures organize data hierarchically or through interconnected relationships rather than a strict sequence. Structures such as trees, graphs, and hash tables belong to this category because their elements can branch, connect, or relate to multiple other elements simultaneously.

The Importance of Data Structures

Data structures are essential because they directly influence how efficiently a program stores, processes, and retrieves information. Properly chosen data structures can drastically improve performance and reduce unnecessary processing time.

- One important role of data structures is **efficient data management**. By organizing data effectively, programs can retrieve and manipulate information much faster. This becomes especially important in applications handling millions of records or real-time systems.
- Data structures also provide **logical organization**. Well-structured information is easier for both humans and computers to understand, maintain, and process. Instead of chaotic collections of values, data becomes arranged in meaningful and manageable forms.
- Another important concept is **data abstraction**. Data structures hide complex implementation details and allow programmers to focus on solving logical problems instead of worrying about low-level memory handling. This abstraction improves productivity and code readability.
- Reusability is another major advantage. Many common data structures are standardized and widely used across applications. Instead of reinventing solutions repeatedly, programmers can reuse proven structures such as stacks, queues, linked lists, and trees.
- Finally, data structures play a crucial role in **algorithm optimization**. The efficiency of an algorithm is often highly dependent on the data structure chosen. A poorly selected structure may cause unnecessary delays and memory consumption, while the correct choice can dramatically improve performance.

Data Structures in Programming Languages

Learning data structures is closely tied to learning programming languages. Although programming languages may differ in syntax, the underlying algorithms and data structure concepts remain largely the same because they aim to achieve identical outcomes. Whether using Python, Java, C, or C++, programmers must not only write functional code but also select suitable data structures and algorithms to solve problems effectively.

A strong understanding of data structures also strengthens problem-solving skills. Complex problems can often be broken down into smaller, manageable steps once the programmer understands how data should be organized and manipulated.

Choosing the correct data structure becomes even more important in situations where processing power or memory is limited. Efficient structures ensure that resources are used wisely while still maintaining acceptable performance. Programs designed with strong foundations in data structures and algorithms are also more scalable, meaning they can handle increasingly large datasets and more complicated operations without major performance loss.

Additionally, data structures and algorithms form a universal language shared by programmers worldwide. Regardless of programming language, developers communicate concepts such as stacks, queues, trees, graphs, sorting, or searching using the same fundamental understanding.

The general process of implementing data structures and algorithms into a program can be summarized through several steps:

- **Step 1: Identify the Problem:** The programmer first analyzes the problem carefully by determining the required inputs, expected outputs, constraints, and necessary variables.
- **Step 2: Choose Appropriate Data Structures:** Different problems require different structures. Arrays may work well for simple static collections, while linked lists support dynamic insertion and deletion. Hierarchical problems may require trees or graphs.
- **Step 3: Design the Algorithm:** Once the structure is selected, the programmer designs a sequence of logical steps that manipulates and processes the data effectively.
- **Step 4: Implement the Solution:** The algorithm is translated into actual code while ensuring readability, maintainability, and proper error handling.
- **Step 5: Analyze the Solution:** The final solution must be evaluated based on correctness, time complexity, and space complexity to ensure overall efficiency and reliability.

Applications of Data Structures

Data structures are used extensively across nearly every field of computing.

- One major application is **data storage**, where structures help database systems organize records efficiently for fast retrieval and modification.

- They are also critical in **data exchange** between systems. Network protocols such as TCP/IP rely on structured packets of information to ensure reliable communication between applications and devices.
- Operating systems heavily depend on data structures for **resource and service management**. Linked lists, queues, and trees help manage files, allocate memory, and schedule processes efficiently.
- Modern big data systems also rely on data structures to maintain **scalability**. Distributed storage systems organize and allocate enormous amounts of information across multiple locations while maintaining performance and reliability.

Data structures offer many practical advantages:

- They support efficient storage and retrieval of information.
- They simplify the processing of both small and massive datasets.
- They help reduce execution time and improve overall program performance.
- They provide reliable, pre-designed methods for operations such as insertion, deletion, searching, and updating.
- Many structures, including arrays, stacks, queues, trees, graphs, and linked lists, are already thoroughly tested and optimized, allowing programmers to use them confidently without redesigning solutions from scratch.
- Their reusability saves development time and encourages consistent software design practices.

Summary

This section introduces one of the most important foundations of Computer Science: algorithms and data structures.

- An algorithm is a sequence of logical instructions used to solve problems and produce outputs from given inputs. Data structures, meanwhile, provide organized ways to store and manage data efficiently so algorithms can operate effectively.
- Data structures are generally divided into two major categories: **linear** and **non-linear** structures. They are essential in software development because

they improve efficiency, scalability, and overall problem-solving capability.

Throughout this course, you will gradually explore how algorithms and data structures work together to solve increasingly complex computational problems.

References

https://www.geeksforgeeks.org/dsa/introduction-to-algorithms/

https://www.geeksforgeeks.org/dsa/data-structure-meaning/

https://www.scribbr.com/ai-tools/what-is-an-algorithm/

https://www.ibm.com/think/topics/data-structure

https://www.codecademy.com/article/what-is-dsa-understanding-data-structures-and-algorithms

Next Section:

1. Greedy Approach